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Propositions and Connectives

Logic is concerned with all kinds of reasoning, whether they be legal arguments or mathematical proofs or conclusions in a scientific theory based on a set of hypotheses. An aim of logic is to provide rules by which one can determine whether any particular argument or reasoning is valid (correct). Due to the diversity of their application, these rules, called ***rules of inference***, must be stated in general terms and must be independent of any particular language used in the arguments. The theory of inference is formulated in such a way that we should be able to decide the validity of the argument by following the rules mechanically and independently of our own feelings about the argument.

We require a language to state rules. Natural languages are not suitable for this purpose since they are not precise and are ambiguous. It is therefore necessary to develop a formal language called **object language**. A formal language is one in which the *syntax* is well defined. In order to avoid ambiguity, we use symbols which have been clearly defined in the object language. An advantage of the use of symbols is that they are easy to write and manipulate. Because of the use of symbols, the logic that we shall study is also called **symbolic logic**.

The study of object language requires the use of a natural language (and we choose English in our case). This natural language will then be called ***meta language.***

Propositions:

A ***declarative sentence*** is a sentence that declares a fact. We begin by assuming that the object language contains a set of declarative sentences which cannot be further broken down or analysed into simpler sentences. These are ***primary declarative sentences***. Only those declarative sentences will be admitted in the object languages which have one and only one of two possible values called ***truth values***. The two truth values are ***true*** and ***false*** which are denoted by the symbols T and F respectively. Occasionally T and F are respectively denoted by 1 and 0 .

Since only two possible truth values are admitted, our logic is sometimes called a ***two-valued logic***.

Declarative sentences in the object language are of two types. The first type includes those declarative sentences which are considered to be primitive in the object language. The declarative sentences of the second type are obtained from the primitive declarative sentences by using certain symbols called ***connectives (logical operators***) and certain punctuation marks, such as parentheses to join primitive declarative sentences. In any case, all the declarative sentences to which it is possible to assign one and only of the two truth values are called ***propositions*** (or ***statements***).

A ***proposition*** (or ***statement***) is a declarative sentence that is either true or false, but not both.

Example 1:

(i) Delhi is the capital of India

(ii) $2 + 3 = 4$

(iii) What is it?

(iv) $x + y = z$

(v) This statement is false

(i) and (ii) are declarative sentences and they are propositions. Proposition (i) is true and the proposition (ii) is false. Sentence (iii) is not a proposition since it is not a declarative sentence. Sentence (iv) is not a proposition, since it is neither true nor false. Note that the sentence (iv) can be turned into a proposition if we assign values to the variables. Sentence (v) is not a proposition since we cannot properly assign a definite truth value. If we assign the value true then (v) implies that the sentence (v) is false. On the other hand, if we assign it the value false then (v) implies that the sentence (v) is true. This example illustrates a ***semantic paradox***.

A ***proposition variable*** (or ***statement variable***) is a variable that represents a proposition. The propositional variables are denoted by lowercase letters. The conventional letters used for propositional variables are $p, q, r, s, \ldots$

The area of logic that deals with propositions is called ***propositional calculus*** or ***propositional logic***. It was first developed systematically by the Greek philosopher Aristotle more than 2300 years ago.

A proposition which does not contain any connectives is called a **atomic** (or **primary** or **simple**) ***proposition***. A proposition which contains one or more atomic propositions as well as some connectives is called a ***molecular*** (or ***compound*** or ***composite***) ***proposition***.

Negation:

Let p be a proposition. The negation of p, denoted by $\sim p$ (also denoted by $\neg p$ or $\bar{p}$), is the proposition " it is not the case that p". The proposition $\sim p$ is read as "not p". The truth value of $\sim p$ is the opposite of the truth value of p.

The truth table of $\sim p$ is a tabular form in which the truth values of $\sim p$ are given for arbitrary truth values of p.

Truth table for $\sim p$	
p	$\sim p$
T	F
F	T

Consider the proposition p : Delhi is a city

Then $\sim p$ is the proposition $\sim p$: It is not the case that Delhi is a city

The above proposition can also be written as $\sim p$: Delhi is not a city

Note that the two propositions "It is not the case that Delhi is a city" and "Delhi is not a city" are not identical but we have denoted both of them by $\sim p$. The reason to denote them by the same $\sim p$ is that they mean the same in English (meta language).

Note: A given proposition in object language is denoted by a symbol and it may correspond to several statements in meta language. This multiplicity happens because in a meta language one can express oneself in a variety of ways.

Note that the negation constructs a new proposition from a single existing proposition. We will now introduce connectives (***and***, ***or, if…then*** … and ***if and only if)*** that are used to form new propositions from two or more existing propositions.

Conjunction:

If p and q are propositions, then the conjunction of p and q is the compound proposition "p and q" denoted by $p \wedge q$. The connective **and** is denoted by the symbol $\wedge$.

The compound proposition $p \wedge q$ is true when both p and q are true and is false otherwise.

The truth table of a compound proposition is a tabular form in which the truth values of the compound proposition are given interms of its component parts.

Truth table for $p \wedge q$		
p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Note that in logic the word "but" sometimes is used instead of "and" in a conjunction. (In meta language, there is a difference in meaning between "and" and "but"; we will not be concerned with this nuance here.)

Disjunction:

If p and q are propositions, the disjunction of p and q is the compound statement "p **or** q" denoted by $p \vee q$. The connective **or** is denoted by the symbol $\vee$.

The compound proposition $p \vee q$ is true if at least one of p or q is true and is false otherwise.

Truth table for $p \vee q$		
p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Note: In English the word "or" is commonly used in two distinct ways. It is used (i) in the *inclusive* sense, i.e., "p or q or both", and (ii) in the *exclusive* sense, i.e., "p or q but not both". For example, "He will go to Harvard University or Yale University" uses "or" in the exclusive sense. In logic $p \vee q$ always means "p and/or q".

Example 2:

Let p and q be propositions given by

p : Sam is rich $\qquad$ q : Sam is happy

Write each of the following in symbolic form:

a. Sam is poor but happy.
b. Sam is either rich or unhappy.
c. Sam is neither rich nor happy.

Solution:

a. $\sim p \wedge q$

b. $p \vee \sim q$
c. $\sim p \wedge \sim q$

Conditional propositions:

If p and q are propositions, then the compound statement "if p, then q" ("if p, q") denoted by $p \rightarrow q$, is called a ***conditional proposition*** or ***implication***.

In $p \rightarrow q$, the proposition p is called ***antecedent*** (or ***hypothesis***) and the proposition q is called ***consequent*** (or ***conclusion***). The connective ***if … then …*** is denoted by $\rightarrow$.

The proposition $p \rightarrow q$ has a truth value F when q has truth value F and p has truth value T and has truth value T otherwise.

Truth table for $p \rightarrow q$		
p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Note:

(i) According to the definition, it is not necessary that there be any kind of relation between p and q in order to form $p \rightarrow q$.
(ii) It is customary to represent any one of the following expressions by $p \rightarrow q$:

q is necessary for p	q when p
p is sufficient for q	a necessary condition for p is q
q if p	q unless $\sim p$
p only if q	a sufficient condition for q is p
q whenever p	q follows from p

(iii) In mathematics "if p then q" and "p implies q" are used interchangeably, but in this text we use the word implies in different way.

The **converse** of the implication $p \rightarrow q$ is the implication $q \rightarrow p$.

The **inverse** of the implication $p \rightarrow q$ is the implication $\sim p \rightarrow \sim q$ (read as if not p then not q).

The **contra positive** of the implication $p \rightarrow q$ is the implication $\sim q \rightarrow \sim p$.

Truth tables for converse, inverse and contrapositive of $p \longrightarrow q$

p	q	$Conditional$ $p \longrightarrow q$	$Converse$ $q \longrightarrow p$	$\sim p$	$\sim q$	$Inverse$ $\sim p \longrightarrow \sim q$	$Contrapositive$ $\sim q \longrightarrow \sim p$
T	T	T	T	F	F	T	T
T	F	F	T	F	T	T	F
F	T	T	F	T	F	F	T
F	F	T	T	T	T	T	T

Example 3:

Write the converse, inverse and contrapositive of the implication "If today is Sunday, then I will go for a walk".

Solution: Let p and q be propositions

p : Today is Sunday $\qquad$ q : I will go for a walk

Then the given proposition is written as $p \longrightarrow q$.

The converse of this implication is

$q \longrightarrow p$: If I will go for a walk, then Today is Sunday

The inverse of the above implication is

$\sim p \longrightarrow \sim q$: If Today is not Sunday then I will not go for a walk

The contrapositive of $p \longrightarrow q$ is

$\sim q \longrightarrow \sim p$: If I will not go for a walk, then Today is not Sunday

Biconditional Propositions:

If p and q are propositions, then the compound statement "p if and only if q", denoted by $\boldsymbol{p \leftrightarrow q}$ **(**or $\boldsymbol{p \rightleftarrows q}$), is called a **biconditional proposition**. The connective ***if and only if*** is denoted by the symbol $\leftrightarrow$ (or $\rightleftarrows$).

The proposition $p \leftrightarrow q$ can also be stated as "p is necessary and sufficient condition for q" (or "if p then q, and conversely" or "p iff q")

The proposition $p \leftrightarrow q$ has truth value T whenever both p and q have identical truth values.

Truth table for $p \leftrightarrow q$		
p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Example 4:

Construct the truth tables for $\boldsymbol{p \wedge {\sim}p}$ and $\boldsymbol{p \vee {\sim}p}$

Truth table for $p \wedge {\sim}p$		
p	${\sim}p$	$p \wedge {\sim}p$
T	F	F
F	T	F

Truth table for $p \vee {\sim}p$		
p	${\sim}p$	$p \vee {\sim}p$
T	F	T
F	T	T

P1:

Which of the following are propositions?

a. **2 is an even integer.**
b. **Why should we study discrete structures?**
c. **There is an integer x such that $x^2 = 3$.**
d. **Please be quiet.**
e. **Dogs can fly.**
f. **There will be snow in December.**
g. **What a beautiful evening!**
h. **Get up and do your exercises.**

Solution:

a. This is a declarative sentence and its truth value is T. Therefore, it is a proposition.
b. This is not a declarative sentence. Therefore, it is not a proposition.
c. This is a declarative sentence, since there is no integer x such that $x^2 = 3$. Its truth value is F. Therefore, it is a proposition.
d. This is not a declarative sentence. Therefore, it is not a proposition.
e. This is a declarative sentence and its truth value is F.
f. This is a declarative sentence and it is either true or false but not both. Therefore, it is a proposition.
g. This is not a declarative sentence. Therefore, it is not a proposition.
h. This is not a declarative sentence. Therefore, it is not a proposition.

P2:

If p and q are the following propositions:

p : 2 is an even integer **q : -3 is a negative integer**

then write the following propositions interms of p, q and logical connectives and find their truth values:

a. **2 is an even integer and -3 is a negative integer.**
b. **2 is not an even integer and -3 is a negative integer.**
c. **2 is not an even integer and -3 is not a negative integer.**
d. **If 2 is not an even integer, then -3 is not a negative integer.**
e. **If 2 is an even integer, then -3 is not a negative integer.**
f. **2 is an even integer if and only if -3 is a negative integer.**
g. **2 is not an even integer if and only if -3 is a negative integer.**

Solution:

First, note that the truth value of p is T and the truth value of q is T.

a. $p \wedge q$, T
b. $\sim p \wedge q$, F
c. $\sim p \wedge \sim q$, F
d. $\sim p \longrightarrow \sim q$, T
e. $p \longrightarrow \sim q$, F
f. $p \leftrightarrow q$, T
g. $\sim p \leftrightarrow q$, F

P3:

Write the converse, the inverse and the contrapositive of the following conditional proposition

"The home team wins whenever it is raining"

Solution:

The given proposition can be rewritten as "If it is raining, then the home team wins". It is written in symbolic form as $p \longrightarrow q$ where

p : It is raining $\qquad$ q : The home team wins

The converse of $p \longrightarrow q$ is $q \longrightarrow p$, i.e., "If the home team wins, then it is raining".

The inverse of $p \longrightarrow q$ is $\sim p \longrightarrow \sim q$, i.e., "If it is not raining, then the home team does not win".

The contrapositive of $p \longrightarrow q$ is $\sim q \longrightarrow \sim p$, i.e., "If the home team does not win, then it is not raining".

P4:

Construct the truth tables for $\sim p \vee q$ and $\sim p \vee \sim q$.

Solution:

Truth table for $\sim p \vee q$			
p	q	$\sim p$	$\sim p \vee q$
T	T	F	T
T	F	F	F
F	T	T	T
F	F	T	T

Truth table for $\sim p \vee \sim q$				
p	q	$\sim p$	$\sim q$	$\sim p \vee \sim q$
T	T	F	F	F
T	F	F	T	T
F	T	T	F	T
F	F	T	T	T

P5:

Construct the truth tables for $\sim p \wedge q$ and $p \wedge \sim q$.

Solution:

Truth table for $\sim p \wedge q$			
p	q	$\sim p$	$\sim p \wedge q$
T	T	F	F
T	F	F	F
F	T	T	T
F	F	T	F

Truth table for $p \wedge \sim q$			
p	q	$\sim q$	$p \wedge \sim q$
T	T	F	F
T	F	T	T
F	T	F	F
F	F	T	F

P6:

Determine the truth values of the following propositions:

a. If Paris is in France then $2 + 2 = 4$
b. If Paris is in France then $2 + 2 = 5$
c. If Paris is in England then $2 + 2 = 4$
d. If Paris is in England then $2 + 2 = 5$

Solution:

Note that the proposition "If p, then q" is false only when p is true and q is false. Therefore, the truth values of the above are

a. T b. F c. T d. T

P7:

Determine the truth values of the following propositions:

a. Paris is in France if and only if $2 + 2 = 4$
b. Paris is in France if and only if $2 + 2 = 5$
c. Paris is in England if and only if $2 + 2 = 4$
d. Paris is in England if and only if $2 + 2 = 5$

Solution:

The propositions (a) and (d) are true since the primary (atomic) propositions are both true in (a) and both false in (d). On the other hand (b) and (d) are false since their atomic propositions have opposite truth values.

P8:

Let p, q be primitive propositions for which the implication $p \longrightarrow q$ is false. Determine truth values for

a. $p \wedge q$
b. $\sim p \vee q$
c. $q \longrightarrow p$
d. $\sim q \longrightarrow \sim p$
e. $p \leftrightarrow q$
f. $\sim p \longrightarrow \sim q$

Answers:

Note that the proposition "If p then q" is false only when p is true and q is false. Therefore, the truth values of the above are

a. F
b. F
c. T
d. F
e. F
f. T

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Propositions and Connectives

Exercise:

1. Which of the following sentences are propositions? What are the truth values of those that are propositions?
    a. Kharagpur is in Andhra Pradesh
    b. Answer this question
    c. $5 + 8 = 13$
    d. $x + 2 = 9$

2. What is the negation of each of these propositions?
    a. Today is Monday
    b. There is no pollution in Jaipur
    c. $2 + 5 = 7$

3. Let p and q be propositions, where

    p : It is below freezing $\qquad$ q : It is snowing

Write the following propositions using p and q and connectives:

a. It is below freezing and snowing
b. It is below freezing but not snowing
c. It is not below freezing and it is not snowing
d. It is either snowing or below freezing (or both)
e. If it is below freezing, it is also snowing
f. That it is below freezing is necessary and sufficient for it to be snowing

4. Let p and q be propositions, where

    p : Swimming at the New Jersey shore is allowed
    q : Sharks have been spotted near the shore

Express each of these compound propositions as an English sentence.

a. $\sim q$
b. $p \wedge q$
c. $\sim p \vee q$
d. $p \longrightarrow \sim q$
e. $\sim q \longrightarrow p$
f. $\sim p \longrightarrow \sim q$
g. $p \leftrightarrow \sim q$

5. Determine whether the following conditional propositions are true of false.
   a. If $1 + 1 = 2$ then $2 + 2 = 5$
   b. If $1 + 1 = 3$ then $2 + 2 = 4$
   c. If $1 + 1 = 3$ then $2 + 2 = 5$
   d. If monkey can fly then $1 + 1 = 3$

6. Determine whether the following biconditionals are true or false:
   a. $2 + 2 = 4$ if and only if $1 + 1 = 2$
   b. $1 + 1 = 2$ if and only if $2 + 3 = 4$
   c. $1 + 1 = 3$ if and only if monkeys can fly
   d. $0 > 1$ if and only if $2 > 1$

7. Construct the truth tables for following:
   a. $p \vee \sim q$
   b. $\sim p \wedge \sim q$

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Propositions and Connectives

Answers:

1.
    a. Prop, F
    b. Not a prop (since it is not a declarative sentence)
    c. Prop, T
    d. Not a prop, since it is neither true nor false

2.
    a. Today is not Monday
    b. There is pollution in Jaipur
    c. $2 + 5 \neq 7$

3.
    a. $p \wedge q$
    b. $p \wedge \sim q$
    c. $\sim p \wedge \sim q$
    d. $p \vee q$
    e. $p \longrightarrow q$
    f. $p \leftrightarrow q$

4.
    a. Sharks have not been spotted near the shore
    b. Swimming at the New Jersey shore is allowed and sharks have been spotted near the shore
    c. Swimming at the New Jersey shore is not allowed or sharks have been spotted near the shore
    d. If swimming at the New Jersey shore is allowed then sharks have not been spotted near the shore

e. If sharks have not been spotted near the shore then swimming at the New Jersey shore is allowed
f. If swimming at the New Jersey shore is not allowed then sharks have not been spotted near the shore
g. Swimming at the New Jersey shore is allowed if and only if sharks have not been spotted near the shore.

5.

a. F
b. T
c. T
d. T

6.

a. T
b. F
c. T
d. F

7.

a.

Truth table for $p \vee \sim q$			
p	q	$\sim q$	$p \vee \sim q$
T	T	F	T
T	F	T	T
F	T	F	F
F	F	T	T

b.

Truth table for $\sim p \wedge \sim q$				
p	q	$\sim p$	$\sim q$	$\sim p \wedge \sim q$
T	T	F	F	F
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T